Thomas Chisica Londoño

Applied Mathematics & Computer Science

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Research Interests

Reinforcement learning, computational neuroscience, and reliable AI, including LLM-based systems. My work combines mathematical modelling, simulation, and controlled experiments.

Education

2024–present	BSc in Applied Mathematics and Computer Science at **Universidad del Rosario**, Bogotá. GPA: 4.11/5.00. Expected graduation: June 2028.
Coursework	Machine Learning (5.0/5.0), Optimisation, Algorithms and Data Structures, Linear Algebra, Probability and Statistics, Graph Theory, Real Analysis, Differential Equations.

Research Experience

Supply-chain control with reinforcement learning 2025–present

Undergraduate research with Prof. Alexander Garrido, Universidad del Rosario

- Rebuilt a thirteen-operation supply-chain model from a thesis as a SimPy simulation, then implemented Gymnasium environments and PPO experiments in Python/PyTorch.
- Compared learned controllers with static policies under matched decision constraints. Stronger baselines removed the apparent PPO advantage, leading to a narrower interpretation of the results.
- Added automated tests, repeatable evaluation scripts, and checks against the source model.

SCRES: code & experiments

Collective behaviour and multi-agent learning 2025–present

Undergraduate research with Prof. Edgar Andrade-Lotero, Universidad del Rosario

- Independently reanalysed a human collective-search experiment using spectral graph methods. Identified a repeated Fiedler eigenvalue that makes a single-eigenvector partition ambiguous, and compared human partitions with symmetry-aware references.
- Evaluated early predictors of later specialisation using leave-one-out cross-validation on 29 dyads; the combined model reached an exploratory AUC of 0.86.

Spectral analysis

Computational models of animal social networks 2026–present

Research collaboration with Dr. Adriana Maldonado-Chaparro, Universidad del Rosario

- Built a Python reference implementation and reproduced runs in NetLogo 5.2 to trace logic errors in an inherited individual-based model of marmot social networks.
- Developed data checks and analysis tables for longitudinal observations, distinguishing group membership from opportunities to observe social interactions.

Research Training

Neuromatch Academy: Computational Neuroscience July 2026

68-hour interactive course; team project and independent follow-up

– Studied connectivity in recurrent networks for a motor task, comparing four densities with eight paired seeds and held-out evaluation in the primary experiment.
– Identified that density also changed the number of trainable recurrent weights. Ran a follow-up holding that number fixed; the original density trend did not recur under this control.

Motor RNN: experiments & controls

Professional Experience

Lead in Emerging Technologies June–December 2025

Digital Business Ventures; assigned to Witty with Progresión

– Built a pipeline to process customer-support emails and recurring questions into a knowledge base for Witty, a financial-services platform used by Progresión clients.
– Developed an LLM-based support chatbot using FastAPI, FAISS and multi-provider embeddings, with heuristic answer-or-escalate routing. Designed it to handle routine queries at low inference cost.

Projects

HeliOS: educational RISC-V kernel GitHub

Contributed to a collaborative C kernel with timer-driven scheduling, a UART shell, synchronisation primitives, and heap allocation. The repository includes QEMU/OpenSBI smoke tests.

ChaosLab: double-pendulum simulation GitHub

Built an interactive Python/Streamlit simulation for exploring nonlinear dynamics and sensitivity to initial conditions.

Teaching

July 2024–July 2025	**Calculus I Teaching Assistant**, Universidad del Rosario. Led problem-solving sessions and supported first-year students.

Awards & Funding

2024	**Andrés Bello National Distinction.** Top 33 nationally in Colombia's 2023 secondary-school examination (ICFES Saber 11).
2024–present	**Merit scholarship covering 100% of tuition**, Universidad del Rosario.

Skills

Programming	Python, C, R, SQL.
ML & modelling	PyTorch, Gymnasium, SimPy, NumPy/SciPy, NetworkX.
Development	Git, Linux, pytest, continuous integration, FastAPI, LaTeX.
Languages	Spanish (native); English (B2 accredited by Universidad del Rosario, 2025).